Inspiring Innovative Ideas

COLLEGE OF ENGINEERING TRIKARIPUR

(Under CAPE , Estd . By Govt . of Kerala)

An ISO 9001-2015 Certified Institution. Accredited By NAAC With 'A' Grade

DEPARTMENT OF COMPUTER SCIENCE

FOUNDATIONS OF COMPUTING: FROM HARDWARE ESSENTIALS TO WEB DESIGN

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Module 1

Computer Hardware – CPU,

Memory - Memory hierarchy: registers, cache, RAM, virtual memory,

Motherboard - Computer Peripherals - I/O devices,

Storage devices- HDDs, SSDs, optical drives,

I/O communication and device management,

Interface cards – Buses – Firmware - Boot process

COMPONENTS OF THE COMPUTER SYSTEM

- There are three components required for the implementation of a computerized input-process-output model:

1. The computer **hardware**, which provides the physical mechanisms to input and output data, to manipulate and process data, and to electronically control the various input, output, and storage components.
2. The **software**, both application and system, which provides instructions that tell the hardware exactly what tasks are to be performed and in what order.
3. The **data** that is being manipulated and processed. This data may be numeric, it may be alphanumeric, it may be graphic, or it may take some other form, but in all cases it must be representable in a form that the computer can manipulate.

- The communication component, which consists of hardware and software that transport programs and data between interconnected computer systems.

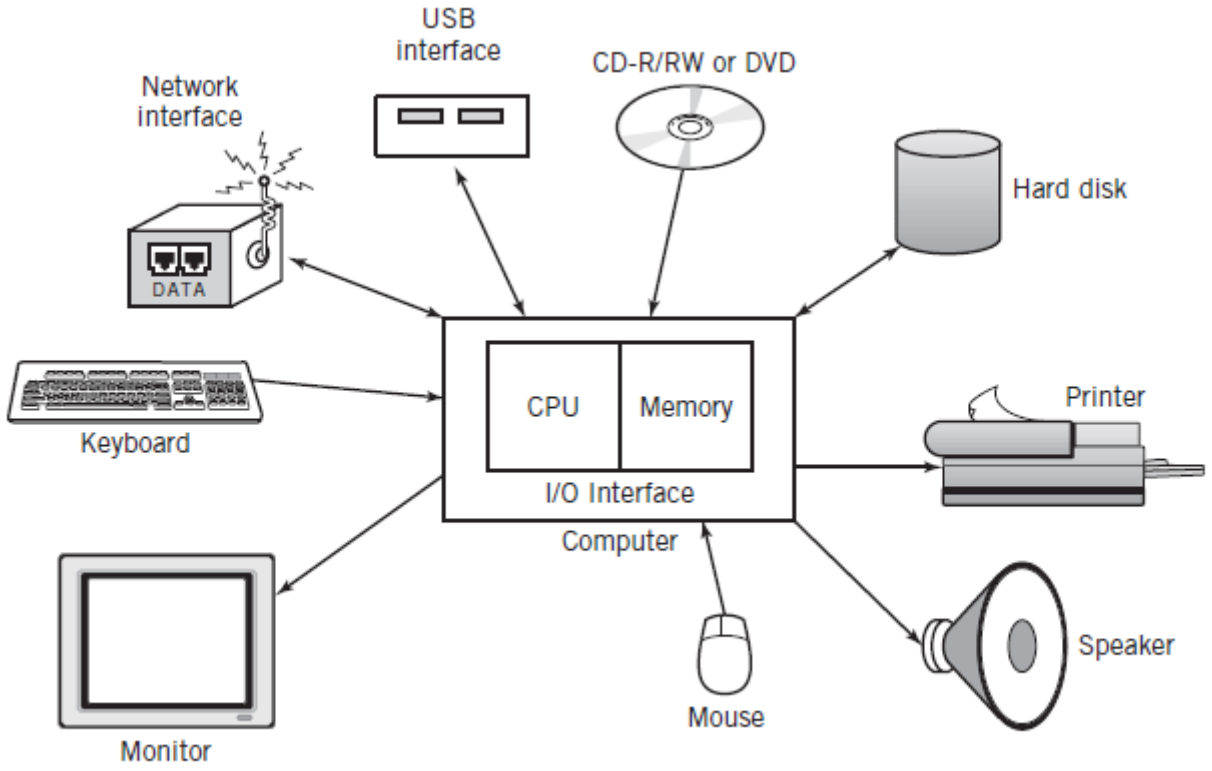
- The hardware and system software components make up the architecture of the computer system. The communication component connects individual computer systems together.

The Hardware Component

- The most visible part of the computer system is obviously the hardware that makes up the system.
- Eg: Consider the computer system upon which you write and execute your programs. You use a keyboard and mouse to provide **input** of your program text and data, as well as for commands to the computer.
- A display screen is commonly used to observe **output**. These are all physical components.
- Calculations and other operations in your program are performed by a **central processing unit (CPU)** inside the computer.
- **Memory** is provided to hold your programs and data while processing is taking place.
- Other input and output devices, such as a disk and SD plug-in cards, are used to provide long-term storage of your program and data files.

- Data and programs are transferred between the various input/output devices and memory for the CPU to use.
- The CPU, memory, and all the input, output, and storage devices form the **hardware** part of a computer system.
- A typical hardware block diagram for a computer is seen in Figure.

Hardware block diagram



Common Input/Output Devices

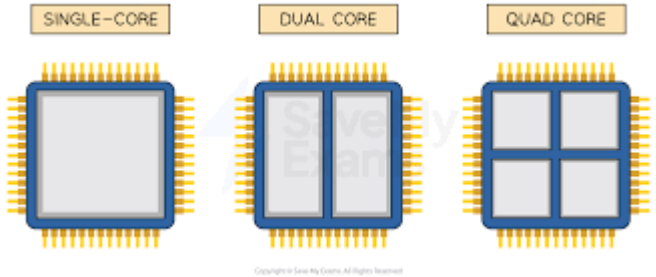
- Bar Code Scanners
- Optical Character Recognition Scanners
- Image Scanners
- RFID Readers
- Video and Audio Capture Devices
- TV Tuners
- Video Cameras
- SD, SmartCard, etc. Card Readers
- Fingerprint and Face Readers
- Touch Screens
- Graphics Tablets
- X-Y Plotters

Central Processing Unit (CPU)

- CPU [Central Processing Unit]. It is the brain of the computer. It is the part that does most of the work in a computer system.
- All types of data processing operations from simple arithmetic to complex tasks and all the important functions of a computer are performed by the CPU.
- A CPU can be installed into a CPU socket. These sockets are generally located on the motherboard.
- **Types of CPU**
- We have three different types of CPU:
- **Single Core CPU:** The oldest type of computer CPU is a single-core CPU. This means that the single-core CPU can only process one operation at a single time. single-core CPU CPU is not suitable for multitasking.

Central Processing Unit (CPU)….

- **Dual-Core CPU:** Dual-Core CPUs contain a single Integrated Circuit with two cores.
- Each core has its cache and controller.
- These controllers and cache work as a single unit. dual-core CPUs can work faster than single-core processors.
- **Quad-Core CPU:** Quad-Core CPUs contain two dual-core processors present within a single integrated circuit (IC) or chip.
- A quad-core processor contains a chip with four independent cores.
- Quad Core CPU increases the overall speed of programs.



Central Processing Unit (CPU)….

The CPU itself is often viewed as a composition of three primary subunits:

1. The **arithmetic/logic unit (ALU)** where arithmetic and Boolean logical calculations are performed.

2. The **control unit (CU)**, which controls the processing of instructions and the movement of internal CPU data from one part of the CPU to another.

3. The **interface unit,** which moves program instructions and data between the CPU and other hardware components.

- The interface unit interconnects the CPU with memory and also with the various I/O (input/output) modules. It can also be used to connect multiple CPUs together.
- In many computer systems, a bus interconnects the CPU, memory, and all of the I/O components.

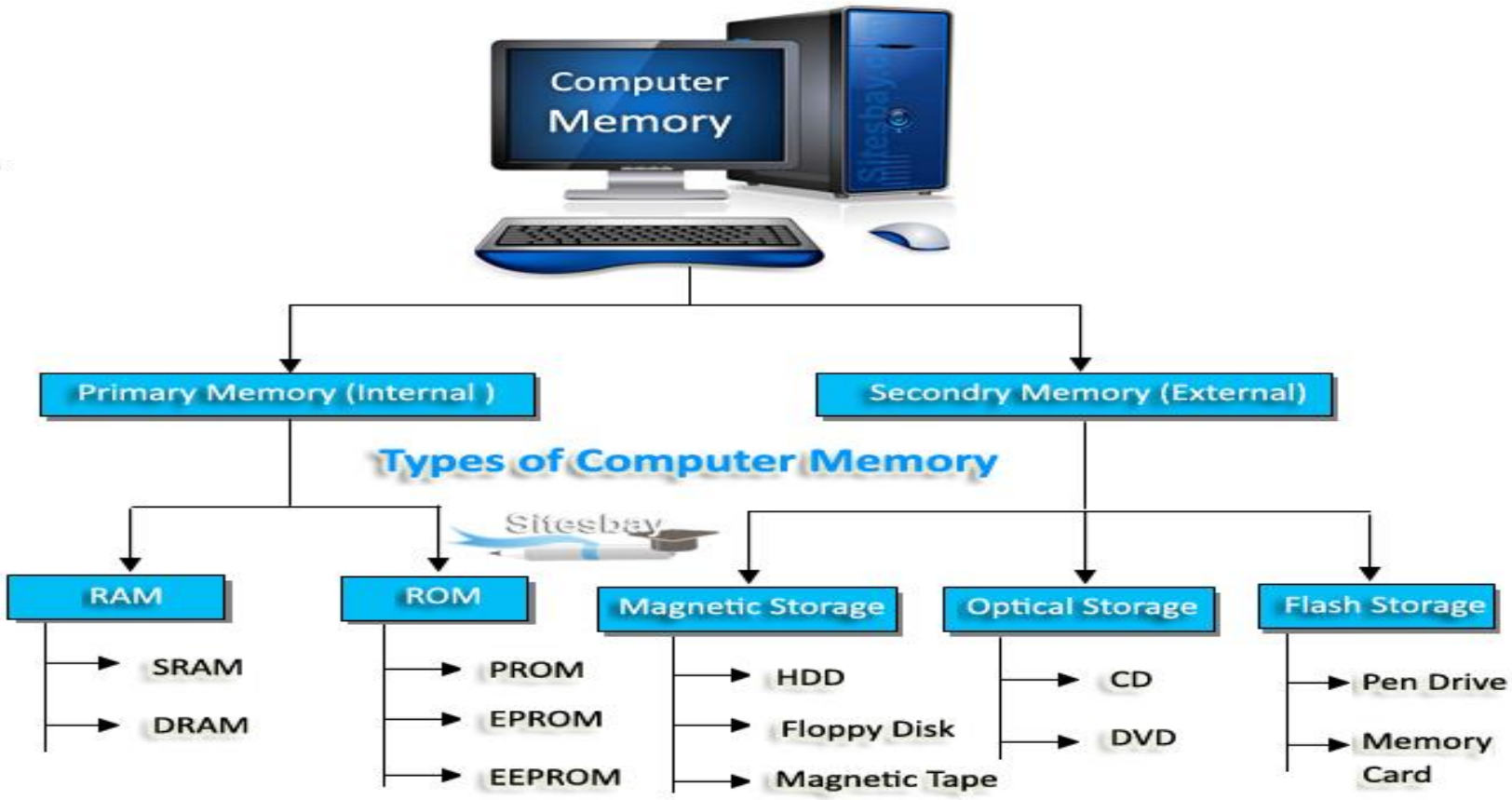
Central Processing Unit (CPU)….

- **A bus** is simply a bundle of wires that carry signals and power between different components.
- In other systems, the I/O modules are connected to the CPU through one or more separate processors known as **channels.**

Memory

- Computer memory is just like the human brain. It is used to store data/information and instructions.
- It is a data storage unit or a data storage device where data is to be processed and instructions required for processing are stored.
- It can store both the input and output can be stored here.
- The main memory, often known as primary storage, working storage, or **RAM** (for **r**andom **a**ccess **m**emory), holds programs and data for access by the CPU.
- **Primary storage** is made up of a large number of cells, each numbered and individually addressable.
- Each cell holds a single binary number representing part of a data value or part of an instruction.
- The smallest addressable size of the cell in most current computers is 8 bits, known as a **byte** of memory.

Memory(Cont..)


Computer Memory
Primary Memory (Internal)
Secondry Memory (External)
Types of Computer Memory
Sitesbay
RAM
ROM
Magnetic Storage
Optical Storage
Flash Storage
SRAM
DRAM
PROM
EPROM
EEPROM
HDD
Floppy Disk
Magnetic Tape
CD
DVD
Pen Drive
Memory Card

Primary Memory

- It is also known as the main memory of the computer system. It is used to store data and programs or instructions during computer operations. It uses semiconductor technology and hence is commonly called semiconductor memory. Primary memory is of two types:
- **RAM (Random Access Memory)**
- **ROM (Read Only Memory)**

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RAM (Random Access Memory):

- The most common type of memory is called random access memory (RAM). As a result, the term *memory* is typically used to mean RAM..
- RAM is like an electronic scratch pad inside the computer. RAM holds data and program instructions while the CPU works with them.
- When a program is launched, it is loaded into and run from memory. As the program needs data, it is loaded into memory for fast access.
- As new data is entered into the computer, it is also stored in memory but only temporarily.
- RAM is volatile, meaning that it loses its contents when the computer is shut off or if there is a power failure. Therefore, RAM needs a constant supply of power to hold its data..

RAM(Cont..)

- The most common measurement unit for describing a computers memory is the byte—the amount of memory it takes to store a single character such as a letter of the alphabet or a numeral.

RAM is of two types:

- **S RAM (Static RAM):** S RAM uses transistors and the circuits of this memory are capable of retaining their state as long as the power is applied.
- This memory consists of the number of flip flops with each flip flop storing 1 bit. It has less access time and hence, it is faster.
- **D RAM (Dynamic RAM):** D RAM uses capacitors and transistors and stores the data as a charge on the capacitors.
- They contain thousands of memory cells. It needs refreshing of charge on capacitor after a few milliseconds. This memory is slower than S RAM.

ROM(Read-Only Memory)

- Unlike RAM, Read-only memory (ROM) permanently stores its data, even when the computer is shut off.
- ROM is called nonvolatile memory because it never loses its contents.
- Whenever the computer's power is turned on, it checks ROM for directions that help it start up, and for information about its hardware devices.
- ROM is used to store information that is used to operate the system. As its name refers to read-only memory, we can only read the programs and data that is stored on it.
- The information stored in the ROM in binary format. It is also known as permanent memory. ROM is of four types:

ROM (Cont..)

- **PROM (Programmable Read Only Memory):** This read-only memory is modifiable once by the user. The user purchases a blank PROM and uses a PROM program to put the required contents into the PROM. Its content can't be erased once written.
- **EPROM (Erasable Programmable Read Only Memory):** EPROM is an extension to PROM where you can erase the content of ROM by exposing it to Ultraviolet rays for nearly 40 minutes.
- **EEPROM (Electrically Erasable Programmable Read Only Memory):** Here the written contents can be erased electrically. You can delete and reprogramme EEPROM up to 10,000 times.
- Erasing and programming take very little time, i.e., nearly 4 -10 ms(milliseconds). Any area in an EEPROM can be wiped and programmed selectively.
- **MROM(Masked ROM):** Hard-wired devices with a pre-programmed collection of data or instructions were the first ROMs. Masked ROMs are a type of low-cost ROM that works in this way.

Difference Between RAM and ROM

Difference	Random Access Memory (RAM)	Read Only Memory (ROM)
Data-Retention	RAM is a volatile memory that could store the data as long as the power is supplied.	ROM is a non-volatile memory that the could retain the data even when the power is turned off.
Read/Write	Read and write operations are supported.	Only read operations are supported.
Use	Used to store the data that has to be currently processed by CPU temporarily.	It is typically used to store firmware or microcode, which is used to initialize and control hardware components of the computer.
Speed	It is a high-speed memory.	It is much slower than the RAM.
CPU Interaction	CPU can easily access data stored in RAM.	CPU cannot easily access data stored in ROM.

Difference Between RAM and ROM

Used as/in	CPU Cache , Primary memory.	Firmware, Micro-controllers.
Accessibility	The data stored is easily accessible.	The data stored is not as easily accessible as in the concerning RAM.
Cost	RAM is more costlier than ROM.	ROM is cheaper than RAM.
Chip Size	A RAM chip can store only a few gigabytes (GB) of data.	A ROM chip can store multiple megabytes (MB) of data.
Function	Used for the temporary storage of data currently being processed by the CPU.	Used to store firmware, BIOS, and other data that needs to be retained.

Secondary Memory

- It is also known as auxiliary memory and backup memory. It is a non-volatile memory and used to store a large amount of data or information.
- The data or information stored in secondary memory is permanent, and it is slower than primary memory.
- A CPU cannot access secondary memory directly. The data/information from the auxiliary memory is first transferred to the main memory, and then the CPU can access it.
- Examples: Hard Disk, Floppy Disk, Magnetic Tapes, etc.

Difference between Primary and secondary memory

Primary memory	Secondary memory
The primary memory of a computer is the main memory that is utilized to store data temporarily.	Secondary memory defines to additional storage devices that are utilized to store data permanently.
Primary memory is temporary.	Secondary memory is permanent.
Primary memory is faster than secondary memory because it is directly accessible to the CPU.	Secondary memory is non-volatile, which means it retains data even when the power is off.
Primary memory is directly accessible by Processor/CPU.	Secondary memory is not directly accessible by the CPU.
Nature of Parts of Primary memory varies, RAM-volatile in nature. ROM- Non-volatile.	It's always Non-volatile in nature.
Primary memory is volatile, which means it is wiped out when the computer is turned off.	Since it is non-volatile, data can be retained in case of a power failure.
Primary memory devices are more expensive than secondary storage devices.	Secondary memory devices are less expensive when compared to primary memory devices.

Difference between Primary and secondary memory

The memory devices used for primary memory are semiconductor memories.	The secondary memory devices are magnetic and optical memories.
It can hold data/information currently being used by the processing unit.	It can hold data/information that are not currently being used by the processing unit.
The capacity of primary memory is usually within the range of 16 to 32 GB.	It stores a considerable amount of data and information. The capacity of secondary memory ranges from 200 GB to some terabytes.
Primary memory is also known as Main memory or Internal memory.	Secondary memory is also known as External memory or Auxiliary memory.
It can be accessed by a data bus.	It can be accessed using I/O channels.
Examples: RAM, ROM, Cache memory, PROM, EPROM, Registers, etc.	Examples: Hard Disk, Floppy Disk, Magnetic Tapes, etc.

Memory Hierarchy

The Storage Hierarchy

Increasing storage capacity

Device	*Typical access times*
CPU registers	0.25 nsec
Cache memory (SRAM)	1-10 nsec
Conventional memory (DRAM)	10-50 nsec
Flash memory	120 μsec
Magnetic disk drive	10-50 msec
Optical disk drive	100-500 msec
Magnetic tape	0.5 and up sec

Increasing access times

Memory Hierarchy (Cont..)

- Computer storage is often conceptualized hierarchically, based upon the speed with which data can be accessed.
- The table in Figure shows this hierarchy, together with some typical access times.
- At the top of the hierarchy are the CPU registers used to hold data for the short term while processing is taking place.
- Access to registers is essentially instantaneous, since the registers are actually a part of the CPU.
- Cache memory, if present, is the fastest memory outside the CPU.
- The CPU will always attempt to access current instructions and data in cache memory before it looks at conventional memory.

Memory Hierarchy (Cont..)

- There may be as many as three different levels of cache L1,L2,L3.
- The CPU accesses the data or instruction in conventional memory if cache memory is not present.
- Next in the hierarchy is conventional memory. Both conventional and cache memory are referred to as **primary memory**. Both provide immediate access to program instructions and data by the CPU and can be used for the execution of programs.
- Below the level of conventional memory, storage in the hierarchy is not immediately available to the CPU, is referred to as **secondary storage**, and is treated as I/O.
- Data and programs in secondary storage must be copied to primary memory for CPU access.
- Except for flash memory, access to secondary storage is significantly slower than primary storage.
- Disks and other secondary storage devices are mechanical in nature, and mechanical devices are of necessity slower than devices that are purely electronic.

Memory Hierarchy (Cont..)

- The location of the desired data is usually not immediately accessible, and the medium must be physically moved to provide access to the correct location.
- This requires a *seek time*, the time needed to find the desired location.
- Once the correct data is located, it must be moved into primary memory for use.
- One important advantage of secondary storage, is **its permanence, or lack of volatility**.
- Flash memory uses a special type of transistor that can hold data indefinitely without power.
- The magnetic media used for disk and tape and the optical media used for DVD and CD disks also retain data indefinitely.
- Secondary storage has the additional advantage that it may be used **to store massive amounts of data**.

Memory Hierarchy (Cont..)

- The additional advantage that secondary storage may be used for **offline archiving**, for moving data easily from machine to machine, and for offline backup storage.
- For example a removable hard disk can be used to move large amounts of data between computers.
- Optical disks require little space and can store large amounts of data for archiving and installation purposes, with rapid mounting for retrieval when required.
- Of the various secondary storage components, flash memory and disk devices are the fastest, since data can be accessed randomly.

Memory Hierarchy (Cont..)

This Memory Hierarchy Design is divided into 2 main types:

- **External Memory or Secondary Memory:** Comprising of Magnetic Disk, Optical Disk, and Magnetic Tape i.e. peripheral storage devices which are accessible by the processor via an I/O Module.
- **Internal Memory or Primary Memory:** Comprising of Main Memory, Cache Memory & CPU registers . This is directly accessible by the processor.

Memory Hierarchy (Cont..)

Registers

- Registers are small, high-speed memory units located in the CPU. They are used to store the most frequently used data and instructions. Registers have the fastest access time and the smallest storage capacity, typically ranging from 16 to 64 bits.

Cache Memory

- Cache memory is a small, fast memory unit located close to the CPU. It stores frequently used data and instructions that have been recently accessed from the main memory. Cache memory is designed to minimize the time it takes to access data by providing the CPU with quick access to frequently used data.

Main Memory

- Main memory , also known as RAM (Random Access Memory), is the primary memory of a computer system. It has a larger storage capacity than cache memory, but it is slower. Main memory is used to store data and instructions that are currently in use by the CPU.

Memory Hierarchy (Cont..)

Secondary Storage

- Secondary storage, such as hard disk drives (HDD) and solid-state drives (SSD) , is a non-volatile memory unit that has a larger storage capacity than main memory. It is used to store data and instructions that are not currently in use by the CPU. Secondary storage has the slowest access time and is typically the least expensive type of memory in the memory hierarchy.

Magnetic Disk

- Magnetic Disks are simply circular plates that are fabricated with either a metal or a plastic or a magnetized material. The Magnetic disks work at a high speed inside the computer and these are frequently used.

Magnetic Tape

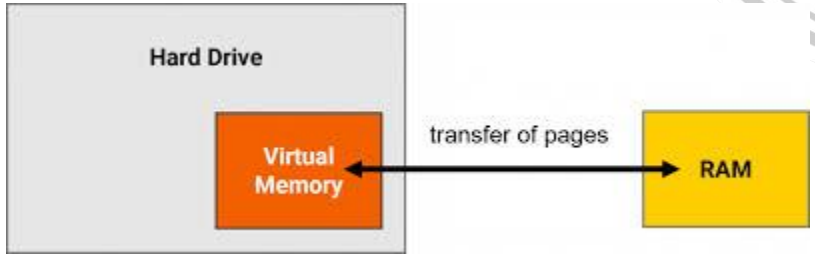
- Magnetic Tape is simply a magnetic recording device that is covered with a plastic film. It is generally used for the backup of data. In the case of a magnetic tape, the access time for a computer is a little slower and therefore, it requires some amount of time for accessing the strip.

Characteristics of Memory Hierarchy

- **Capacity:** It is the global volume of information the memory can store. As we move from top to bottom in the Hierarchy, the capacity increases.
- **Access Time:** It is the time interval between the read/write request and the availability of the data. As we move from top to bottom in the Hierarchy, the access time increases.
- **Performance:** One of the most significant ways to increase system performance is minimizing how far down the memory hierarchy one has to go to manipulate data.
- **Cost Per Bit:** As we move from bottom to top in the Hierarchy, the cost per bit increases i.e. Internal Memory is costlier than External Memory.

Virtual Memory

- Virtual memory is a memory management technique that uses secondary memory to extend the amount of physical memory available on a computer.
- It's a common feature of modern computers' operating systems (OS).
- It allows a computer to compensate for physical memory limitations by temporarily transferring data from RAM to disk storage.



How it works

- Virtual memory uses hardware and software to move data between the computer's RAM and storage devices like hard drives or solid-state drives (SSDs).
- This allows the computer to run programs that require more memory than is physically available.
- Virtual memory can also help with multitasking and improving system performance.

Benefits

- Virtual memory can help programs share memory, which can improve security.
- It can also help optimize CPU usage.

Drawbacks

- Virtual memory can be slower than RAM, which can cause the computer to slow down. This is called thrashing**.**
- Virtual memory can take up storage space that could be used for other purposes

Motherboard

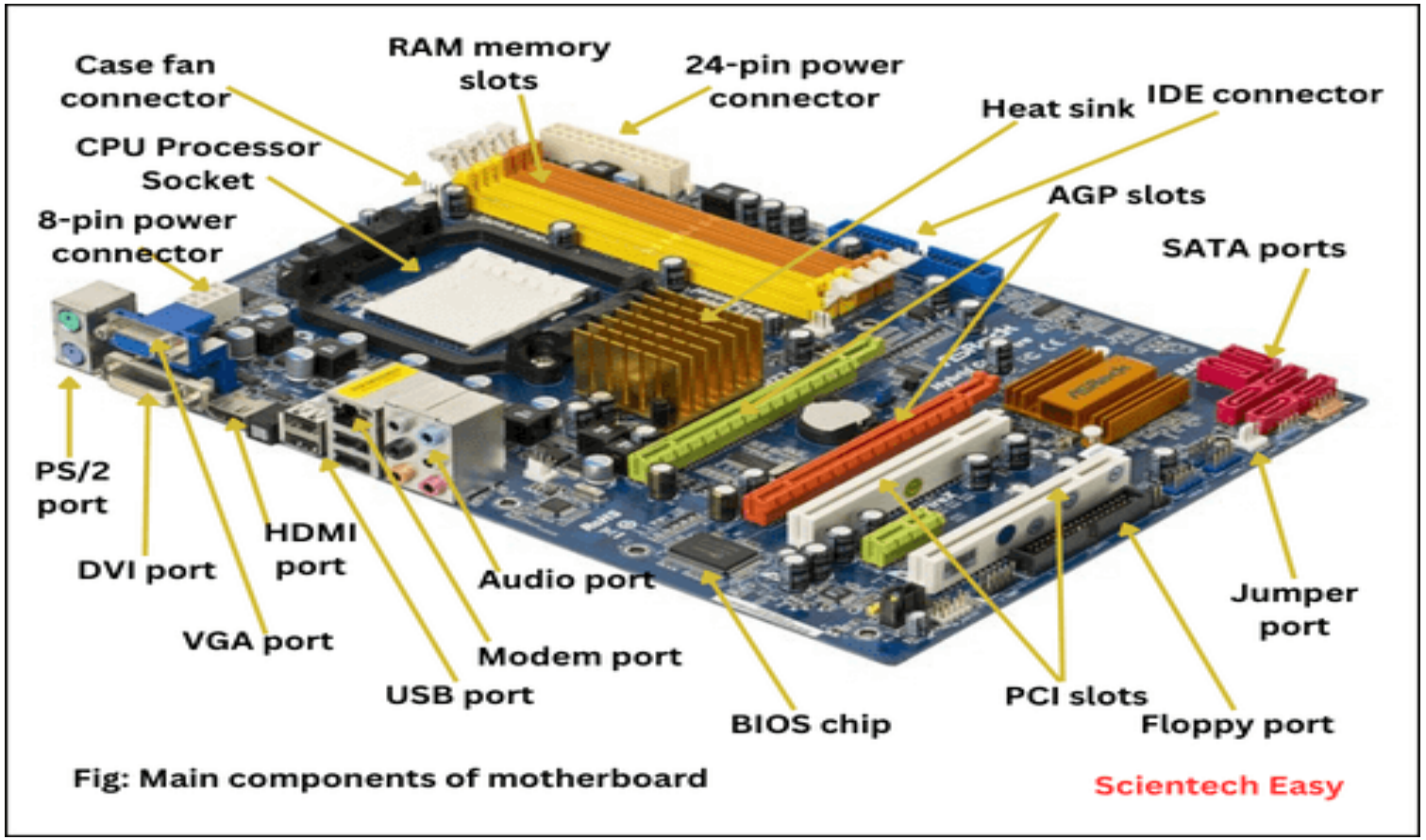


Fig: Main components of motherboard

Motherboard(Cont..)

- A computer is an electronic device that processes the data with respect to the user's requirements using IO devices.
- The data processing takes place in a processor, an important component.
- The processor is situated in a hardware circuit board called the motherboard or printed circuit board (PCB).
- It distributes electricity and facilitates communication between and to the central processing unit (CPU), random access memory (RAM), and any other component of the computer's hardware.
- A computer motherboard may be found inside a computer casing, which is the point of connection for most of the computer's elements and peripherals.

TYPES OF MOTHERBOARD

Advanced Technology (AT) motherboard

- Due to their larger physical dimensions (which can be measured in hundredths of millimeters), these motherboards do not work properly with computers.
- The power connections on these motherboards are in the form of sockets and plugs with six prongs each.
- Due to the difficulty in recognizing these power connections, users often have issues while trying to connect and operate them.

Standard ATX motherboard

- ATX is an enhanced version of the AT motherboard that Intel created in the 1990s. Its name means “advanced technology extended,” and its initials stand for “advanced technology.”
- Unlike AT, it is much more compact and enables the associated components to be interchanged.
- Standard ATX motherboards are ideal for users who need more expansion options, such as adding multiple GPUs, additional storage devices, and PCIe cards for heavy workloads like gaming or video editing.

Micro ATX motherboard

- The length and width of these motherboards, measured in millimeters, are also 244 mm. This motherboard has fewer ports and slots than the Standard ATX board.
- Users who do not want excessive connections and subsequent upgrades, like adding more RAM, an extra GPU, or other Peripheral Component Interconnect (PCI) cards, are better suited for this kind of motherboard than others.

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eXtended ATX motherboard

- The dimensions of this motherboard are 344 millimeters by 330 millimeters (dimensions will differ with different manufacturers). This motherboard supports a single or a twin CPU configuration and has up to eight RAM slots.
- Additionally, it has a higher number of PCIe (where e is for Express) and PCI slots, which may be used to add PCI cards for a wide range of applications.
- EATX motherboards are best suited for professionals running resource-heavy applications, such as 3D rendering, AI modeling, or server management.

Flex ATX motherboard

- Flex ATX is a modification of mini ATX that Intel created between 1999-2000. It is a motherboard standard.
- They were designed to occupy a minimal amount of space and had a minimal price tag.

Low-Profile EXtended (LPX) motherboard

- This has two significant enhancements. The first change was that the output and input ports were moved to the rear of the device, and the second change was the addition of a riser card, which enables the device to have additional slots and makes it easier to attach components.
- The primary drawback of this board is that it does not have any accelerated graphic port (AGP) ports, resulting in a connection to PCI that is made directly.

BTX motherboard

- Balanced technology extended, abbreviated as BTX, is a strategy developed to fulfill the requirements of emerging technologies, which call for increased power consumption and, as a result, emanate more heat.

Pico BTX motherboard

- Given their diminutive size compared to a typical motherboard, these boards are called Pico.
- Even though the upper half of the BTX is shared, support is provided for two expansion slots.

Mini ITX motherboard

- It is important to note that there is no regular-sized version of the information technology extended (ITX) motherboard.
- Due to its reduced power consumption and quicker cooling capabilities, it is primarily used in computers with a small form factor (SFF).
- Given that it has a relatively low level of fan noise, the motherboard is the one that is recommended the most for use in home theater systems because it will enhance the overall performance of the system.

Mini STX motherboard

- The name “Intel 5×5” was initially given to the motherboard now known as the Mini-STX, which stands for mini socket technology extended.
- Although it was introduced in 2015, the motherboard has dimensions of 147 millimeters by 140 millimeters. This converts to a length of 5.8 inches and a width of 5.5 inches; hence, the 5×5 name is rather misleading.

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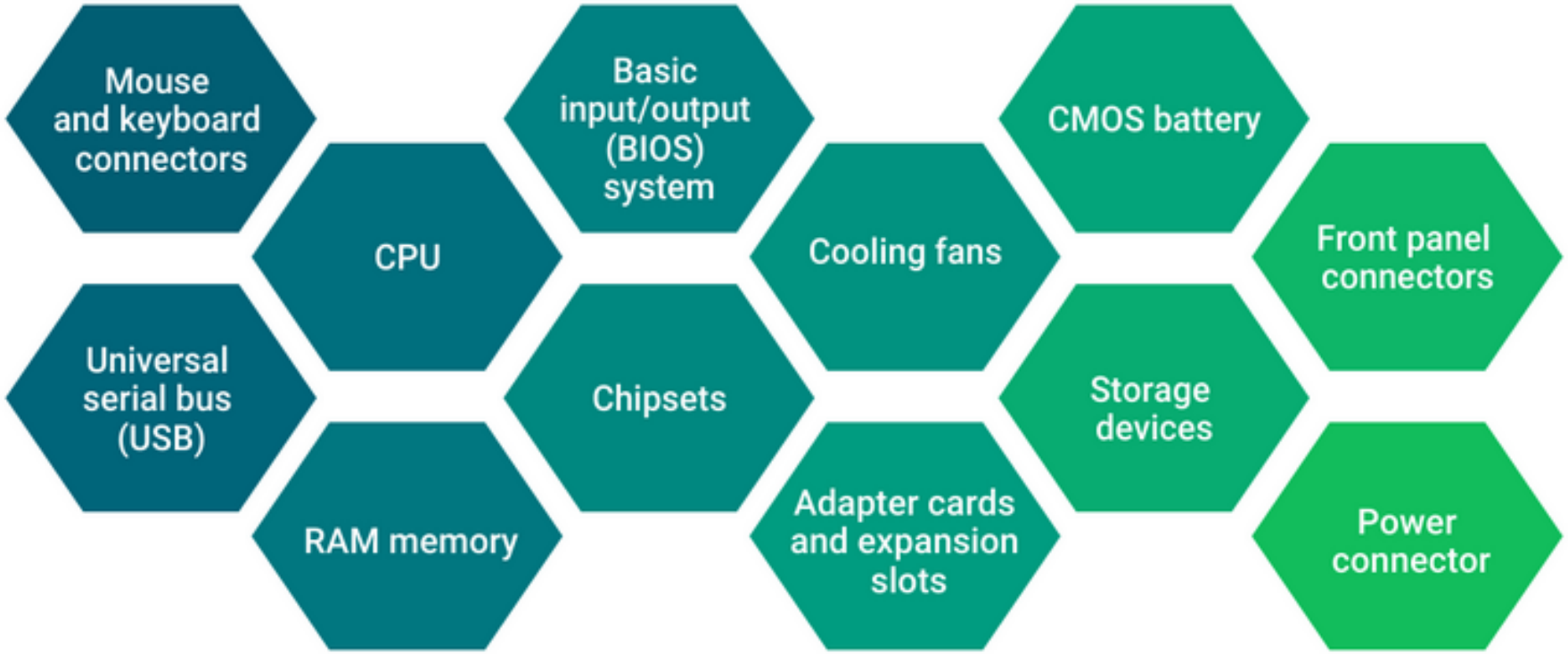
How does a motherboard work?

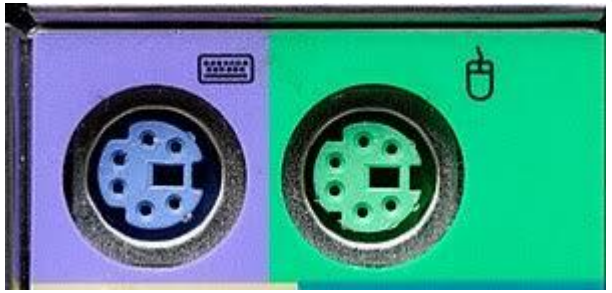
- When you turn your computer on, the power supply transfers electricity to the motherboard to be used by the computer.
- Data is transported between the chipset components via data buses and travels between the southbridge and northbridge sections.
- The data connections to the CPU, RAM, or PCIe are made through the northbridge component.
- The data connection to the basic input/output system (BIOS), the universal serial bus (USB), the serial advanced technology attachment (SATA), and the PCI bus are managed by the southbridge component.
- The information sent over buses will be encoded using a programming language (1 and 0).
- When a signal is sent to a motherboard from one of its components, the motherboard will process it and translate it into a language the other component can comprehend.
- On most of today's computing systems, all of this will occur in a split second, and there is almost no delay between the input and the output.

Key Components of a Motherboard



Mouse and keyboard connectors

- Computer motherboards must have two separate connectors that allow users to connect their external mouse and keyboard.
- There are two keyboard and mouse connectors, the PS/2 and the USB. The personal system/2(PS/2) port is a mini-DIN plug that contains six pins and connects the mouse or keyboard to an IBM-compatible computer. Other computers use the USB port to connect the mouse or keyboard.



Universal serial bus (USB)

- The USB is a computer interface that connects computers to other devices, such as phones.
- The USB port is a significant part of a motherboard that allows users to connect external peripheral devices such as printers, scanners, and pen drives to the computer.
- It enables users to transfer data between the device and the computer.
- Types of USB include USB-A, USB-B, USB-mini, micro-USB, USB-C, and USB-3.

CPU

- The central processing unit (CPU) is commonly referred to as the computer's brain. The CPU controls all the functions of a computer.
- CPUs are available in different form factors, each requiring a particular slot on the motherboard.
- A CPU can contain one or multiple cores. A CPU with a single core can only perform a single task at a time, while those with multiple cores can execute multiple tasks simultaneously.

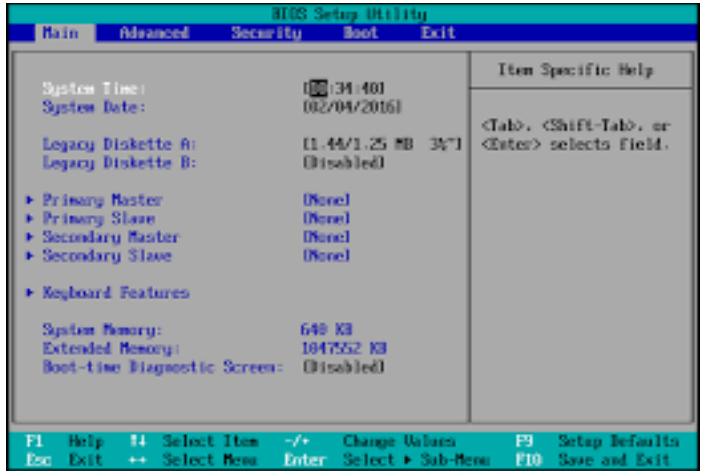
RAM memory

- RAM slots connect the random access memory (RAM) to the motherboard. RAM allows the computer to temporarily store files and programs that are being accessed by the CPU.
- Computers with more RAM capacity can hold and process larger files and programs, thus enhancing performance.
- A computer usually has two RAM slots. However, some computers have up to four RAM slots in the motherboard to increase the available memory.

Basic input/output (BIOS) system

- The BIOS contains the firmware of the motherboard. It consists of instructions about what to do when the computer is turned on.
- It is responsible for initializing the hardware components and loading the computer's operating system.
- The BIOS also allows the computer's operating system to interact and respond with input and output devices such as a mouse and keyboard.
- In some motherboards, the legacy BIOS is replaced by the modern extensible firmware interface (EFI) or the unified extensible firmware interface (UEFI).

- UEFI and EFI allow the computer to boot faster, provide more diagnostic and repair tools, and provide a more efficient interface between the operating system and computer components.



Chipsets

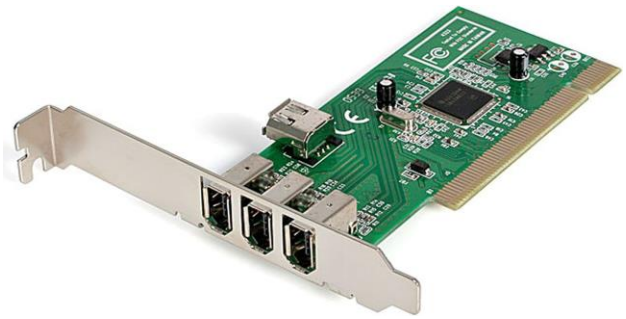
- The chipsets of a computer control how the computer hardware and buses interact with the CPU and other components.
- Chipsets also determine the amount of memory users can add to a motherboard and the type of connectors that the motherboard can have.
- The first type of chipset is the northbridge chipset. The northbridge manages the speed at which the CPU communicates with the components.
- It also controls the processor, the AGP video slot, and the RAM.
- The second type of chipset is the southbridge chipset. The southbridge chipset controls the rest of the components connected to the computer, including communication between the processor and expansion ports such as USB ports and sound cards.

Cooling fans

- The heat generated when electric current flows between components can make a computer run slowly.
- If too much heat is left to build up unchecked, it could damage computer components. Thus, a computer performs better when kept cool.
- Cooling fans increase the airflow, which helps to remove heat from the computer. Some elements, such as video adapter cards, have dedicated cooling fans.

Adapter cards and expansion slots

- Adapter cards are integrated into the motherboard to enhance a computer's functionality. Examples include sound and video adapters.
- The expansion slots allow users to install compatible adapter cards.
- Examples of expansion slots include the peripheral component interconnect (PCI) slot, the AGP slot (which enables the insertion of video cards), the PCI Express serial bus slot, and the PCI-extended slot.

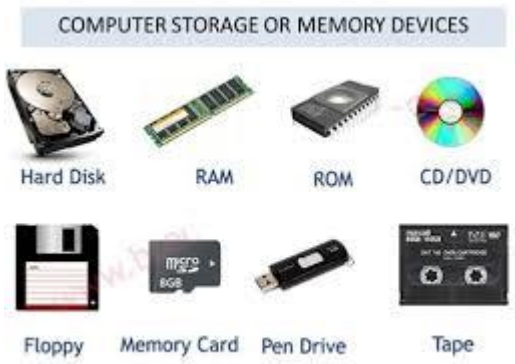


CMOS battery

- The CMOS battery is a small round battery found on the motherboard of every computer.
- It provides power to the complementary metal oxide semiconductor (CMOS) chip. The CMOS chip stores BIOS information and computer settings, even when powered down.
- The CMOS battery allows users to skip resetting BIOS configurations, such as boot order, date, and time settings, each time they power on their computer.

Storage devices

- Storage drives store data permanently or retrieve data from a media disk. The storage devices can either be installed in the computer as hard drives or in removable drives that can connect to the computer through the USB ports.
- Hard disk drives(HDD) or solid-state drives (SSD) are computers' primary storage drives. Computers with SSDs execute tasks much faster and perform better than HDDs.
- Users can also use optical drives such as compact discs to store information.

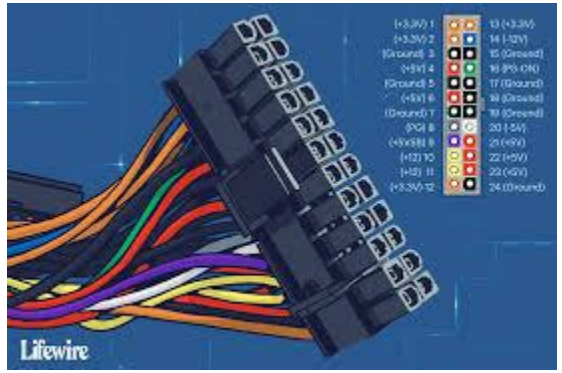


Front panel connectors

- Front panel connectors connect the light-emitting diode (LED) lights on the front of the case to the hard drive, the power button, the reset button, and the internal speaker for testing. Some USB and audio devices also have LED lights.
- These front panel connectors are usually plugged into small pins on the motherboard. Although the pins are grouped and color-coded, their layout structure varies depending on the model of the motherboard.

Power connector

- The power connector provides an electric supply to the computer to function as intended.
- The power supply connector has 20 pins and converts 110-V AC power into +/-12-Volt, +/-5-Volt, and 3.3-Volt direct current (DC) power.



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Northbridge

- It is an integrated circuit within the chipset on the motherboard that is responsible for providing a connection between AGP, CPU interface, and memory.
- Unlike southbridge, it is directly connected to the CPU interface, AGP, and memory. The primary task of northbridge is to provide communication between CPU and external devices via buses.

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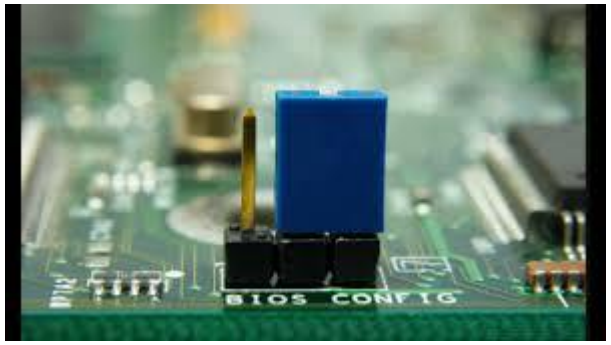
Southbridge

- It is an integrated circuit on the motherboard that is designed for a single function and manufactured as a single unit. It is important for I/O controller, hard drive controller, and integrated hardware.

Jumpers

- It is a tiny metal connector that allows the computer to close an electrical circuit, by which electricity can flow to the certain areas of the circuit board.
- It is an alternative for a dual in-line package switch, and it contains two or more connecting points that control the electrical circuit board.



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Integrated circuit

- It is a small chip that is also known as a monolithic integrated circuit, microchip, or a bare chip.
- It includes multiple circuits, pathways, logic gates, and other components that perform together for a specific function.

PCI slot

- It allows users to add internal components to a desktop computer.
- In 1992, it was designed and introduced by Intel.
- It stands for Peripheral Component Interface that is used to connect the addition of PCI devices such as modems, sound cards, video cards, and network hardware cards.

Serial port connector

- It is an interface that is used to transmit or receive one-bit data at a time on the computer.
- Mostly IBM compatible computers have serial ports as a communication port. For instance, a modem may connect with COM port1 and a mouse connect to COM port2.

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Computer Peripherals - I/O devices

What Does Peripheral Device Mean?

- A **Peripheral Device** is defined as a device that provides input/output functions for a computer and serves as an auxiliary computer device without computing-intensive functionality.

- A peripheral device is also called a peripheral, computer peripheral, input-output device, or I/O device.

Classification of Peripheral devices

- **1. Input Devices:**
- **2. Output Devices:**
- **3. Storage Devices:**
- **4. Communication Devices:**

Input Devices

Keyboard

- The keyboard is the most basic and very commonly used input device which helps to input data to the computer.
- The layout of the buttons in a normally used keyboard is similar to the traditional typewriter, but there are a few additional keys provided by different manufacturers for performing additional functions.
- Normally available keyboards in the market were of two sizes 84 keys and 101/102 keys, but now keyboards with 104 keys or 108 keys are also possible for Windows and Internet.

Mouse(Pointing Device)

- It is the most commonly used pointing device.
- It is a very famous cursor-control device and the earlier versions of it were built over a small palm-size box with a round ball at its base, which tracks the movement of the mouse and feeds digital signals to the CPU when the mouse buttons are pressed.
- Currently, the mouse has a sensor at the bottom to detect cursor position.
- Basic versions if it has two buttons called the left and the right-click button and a wheel is present between the buttons to provide a scroll function.
- The movement of the mouse on a flat surface is used to control the position of the cursor on the display, but it cannot be used to feed text into the system directly.

Joystick

- Just like a mouse, a Joystick is also a pointing device, which is used to travel the cursor position on a display.
- It is a perpendicular stick having a spherical ball at both lower and upper ends.
- The spherical ball connected at the circuit having sensors moves in a socket. The joystick controller can be traversed in all directions.
- The functionality of the joystick is just like a mouse. Its applications are generally in Computer-Aided Designing (CAD) and playing computer games.

Light Pen

- Just like a pen, a light pen is a digital pointing device. This device is used to select an item displayed in the menu or to draw anything on the display.
- It contains a photocell and an optical sensor placed in a small tube at the tip.
- When the light pen tip is traveled on the display, its photocell sensing element calculates the screen location and sends the digital signal to the CPU.

Track Ball

- The trackball is another type of input device similar to the mouse.
- It is most helpful with a notebook or laptop computer, instead of a mouse.
- It consists of a ball on the top which is half inserted and connected to the sensors. By moving fingers on the ball, the cursor can be moved.
- Because the whole device is not moved to move the cursor, a trackball requires less space as compared to the mouse. This device comes in different shapes like a ball, a square, or a button.

Scanner

- A scanner is another input device, which works very much like a photocopy machine.
- It can be used when some information on paper is to be transferred to the hard disk of the computer in digital format for further manipulation.
- The scanner captures high-resolution images from the source which are then converted into a digital format that can be saved on the disk and shared digitally.
- These digital images can be edited before they are printed.

Digitizer

- A digitizer is also an input device that is used to convert analog information into digital form.
- Using a digitizer, the signal from the television or camera can be transformed into a series of numbers that could be saved in a computer hard drive.
- They can be helpful with the computer to create a picture of whatever the camera has been pointed at.
- Digitizer is also called Tablet or Graphics Tablet because it transforms graphics and pictorial data into binary inputs to the system.

Microphone

- A microphone or mic is a very common input device used to input sound that is then stored in a digital form.
- Nowadays microphones are used in almost every industry and devices related to sound recording or transmission.
- The microphone is used for different applications like adding sound to a multimedia presentation or for mixing music, voice calling and recording, video conferencing, and more.

Optical Character Reader (OCR)

- It is an input device that is used to read a printed text similar to the scanner, but the method of reading and the type of output generated is different.
- The format to be fed to the OCR is predefined and can't be used as an ordinary scanner.
- OCR, unlike a scanner, scans the text optically, character by character, converts them into a machine-readable digital code instead of creating high-resolution images, and stores the text on the system memory.

Bar Code Readers

- Bar Code Reader is an input device used to read special format barcoded data (data in the form of light and dark lines).
- Barcoded data is commonly used to create price tags, in labeling goods, numbering the books, etc.
- A flash of light from the device strikes the surface of the object and reflects the sensors present behind the source of light to collect the input.
- Bar Code Reader can only scan a bar code image, converts it into an alphanumeric value, which is then transferred to the computer that the bar code reader is connected to.

Webcam

- A Webcam is a camera that is combined with a computer. Unlike a digital camera, they cannot function independently and should be connected to the computer/laptop for operation.
- It functions as an input device that can take pictures and is mostly used for video streaming and video conferencing or chatting.

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Output Devices

Output Devices

- A piece of hardware/equipment that presents the result of the entered input, once it is processed, is called an output device.
- Several output devices display output in various ways such as images, text, hard copies, and audio/video.

Monitors

- Monitors or Visual Display Unit (VDU), are the primary output devices of a computer system.
- It generates images from tiny dots, termed pixels that are arranged in a well-defined rectangular form.
- The depth and sharpness of the image depend upon the number and size of the pixels.

Printers

- A printer is a very commonly practiced output device, which is used to print information on paper.
- Printers are usually connected to a computer and are used to print documents and images created in word processors, spreadsheets, and other programs.
- A printer receives digital images and text from a computer
- A driver or software converts the file into a language the printer can understand
- The printer uses a series of dots to recreate the image or text on the page
- Types of printers :

1. **Inkjet**: Sprays ink from a cartridge onto the paper
2. **Laser**: Uses a laser beam to attract toner to the paper
3. **Dot matrix**: An impact printer that strikes the paper a line at a time etc

Projector

- A projector is an optical output device that can take images produced by a computer and project or present them onto a screen or surface called a projector.
- Projectors are most commonly used in offices, classrooms, auditoriums, theatres to watch presentations, videos, or pictures generated from a computer or laptop.
- In the older version of projectors, images were created by shining the light through a small transparent lens on the projector; however, the new version of projectors uses laser technology for the same.

Speakers

- Computer speakers are the most popular output devices used with a computer to listen to sound.
- These speakers receive audio as input either in analog or digital form and generate sound as an output.
- With the advancement of technology, speakers are now available with wireless and BlueTooth facilities as well.

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Headphones

- The headphones or earphones or headsets are hardware output tools that are either plugged into the computer or can be wireless.
- They offer the same function as a speaker, the only difference is the frequency of sound.
- With speakers, the sound can be detected over a larger distance while utilizing headphones, the sound is only audible to the individual using them.

Plotters

- A plotter is a computer output device that is primarily used to produce high-quality drawings, designs, and graphics.
- It is different from other printers as it uses one or more pens or markers to draw continuous lines on paper or other suitable materials, rather than using dots or pixels to create images.

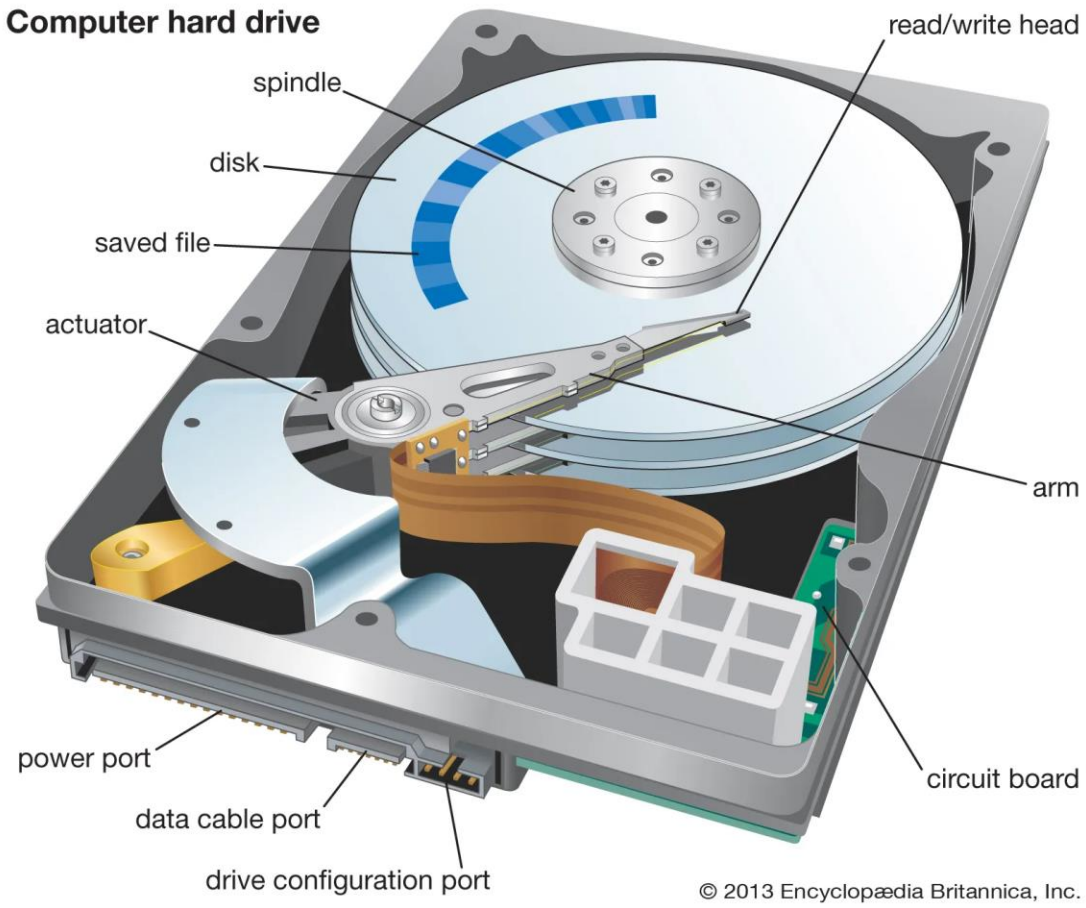
Storage devices- HDDs, SSDs, Optical drives

What is a storage device?

- Storage devices are the computer hardware used to remember/store data.
- There are many types of storage devices, each with their own benefits and drawbacks.

    1. Hard Disk
    2. SSD
    3. Optical Devices etc

Hard Disk Drive (HDD)



Hard Disk Drive (HDD)

- Hard disk drives are non-volatile magnetic storage devices capable of remembering vast amounts of data.
- An electromagnet in the read/write head charges the disk's surface with either a positive or negative charge, this is how binary 1 or 0 is represented.
- The read/write head is then capable of detecting the magnetic charges left on the disk's surface, this is how data is read.

Hard Disk Drive(Cont..)

- A magnetic disk consists of one or more flat, circular platters made of glass, metal, or plastic, and coated with a magnetic substance.
- Particles within a small area of the magnetic substance can be polarized magnetically in one of two directions with an electromagnet; an electromagnet can also detect the direction of polarization previously recorded.
- Thus, magnetic polarization can be used to distinguish 1s and 0s. Electromagnetic read/write heads are used for this purpose.
- A drive motor rotates the disk platter(s) about its central axis. On most drives, the motor rotates the disk at a fixed speed.
- An arm has the read/write head mounted at the end.

Hard Disk Drive(Cont..)

- The arm makes it possible for the head to move radially in and out across the surface of the disk.
- A head motor controls precisely the position of the arm on the disk.
- Most **hard disk drives** contain several platters, all mounted on the same axis, with heads on each surface of each platter.
- The heads move in tandem, so they are positioned over the same point on each surface.
- Except for the top and bottom, each arm contains two read/write heads, which service the surfaces of two adjoining platters.
- With the head in a particular position, it traces out a circle on the disk surface as the disk rotates; this circle is known as a **track**.

Hard Disk Drive(Cont..)

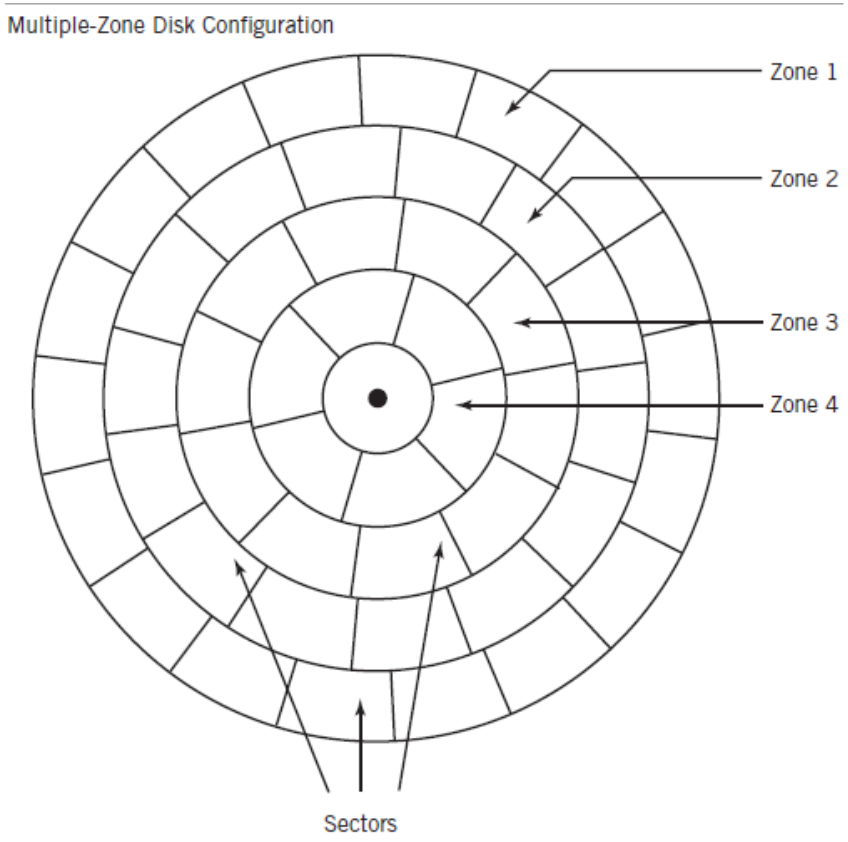
- The heads on each surface all line up, the set of tracks for all the surfaces form a **cylinder**. Each track contains one or more blocks of data.
- On most disks the surface of the disk platter is divided into equally sized pie shape segments, known as **sectors**, although the disks on some large computers divide up the track differently.
- Each sector on a single track contains one **block** of data, typically 512 bytes, which represents the smallest unit that can be independently read or written.
- Regardless of the track, the same angle is swept out when a sector is accessed; thus, the transfer time is kept constant with the motor rotating at a fixed speed. This technique is called **CAV**, for **constant angular velocity**. CAV has the advantage of simplicity and fast access.

Hard Disk Drive(Cont..)

- If there were more bits packed into the outer tracks, the data would be transferred faster at the edge than at the center.
- The motor speed would be adjusted such that the speed *along the track* would be constant regardless of the position of the head. This approach is called **CLV**, for **constant linear velocity**. The capacity of a CLV disk with the same diameter and bit density is approximately double that of an equivalent CAV disk.
- CLV technology is commonly used with CDs and DVDs, the design makes it more difficult to access individual blocks of data rapidly, so it is rarely used for hard disks.

Hard Disk Drive(Cont..)

- Modern disk drives divide the disk into a number of zones, typically sixteen.



Hard Disk Drive(Cont..)

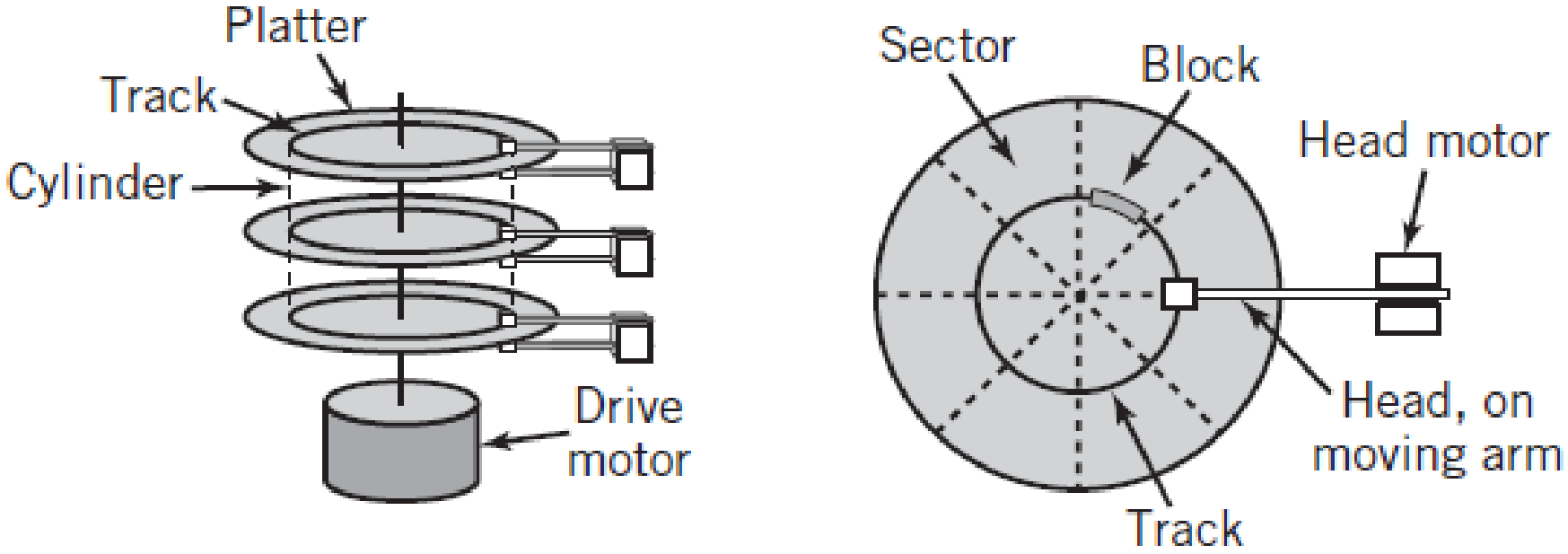
- The largest number of sectors will be in the zone containing the outermost cylinders, Instead of adjusting the motor speed, the disk controller buffers the data rate so that the data rate to the I/O interface is constant, despite the variable data rate between the controller and the disk.
- Different vendors call this technique **multiple zone recording**, **zone bit recording (ZBR)**, or **zone-CAV recording (Z-CAV)**.
- The platter on a hard disk drive is made of a rigid material and is precisely mounted.
- A typical hard disk rotates at 5400 revolutions per minute (rpm), 7200 rpm, or even 10,800 rpm.

Hard Disk Drive(Cont..)

- The disk surface is divided into concentric circles (tracks) and sectors (wedges). Dividing the surface in this way provides physical addresses to remember where data is saved.
- The point to be noted here is that the outer tracks are bigger than the inner tracks but they contain the same number of sectors and have equal storage capacity.
- Typical HDD capacities are measured in Terabytes (TB).
- The read-write (R-W) head moves over the rotating hard disk.
- It is this Read-Write head that performs all the read and write operations on the disk and hence, the position of the R-W head is a major concern.

Hard Disk Drive(Cont..)

A Hard Disk Layout

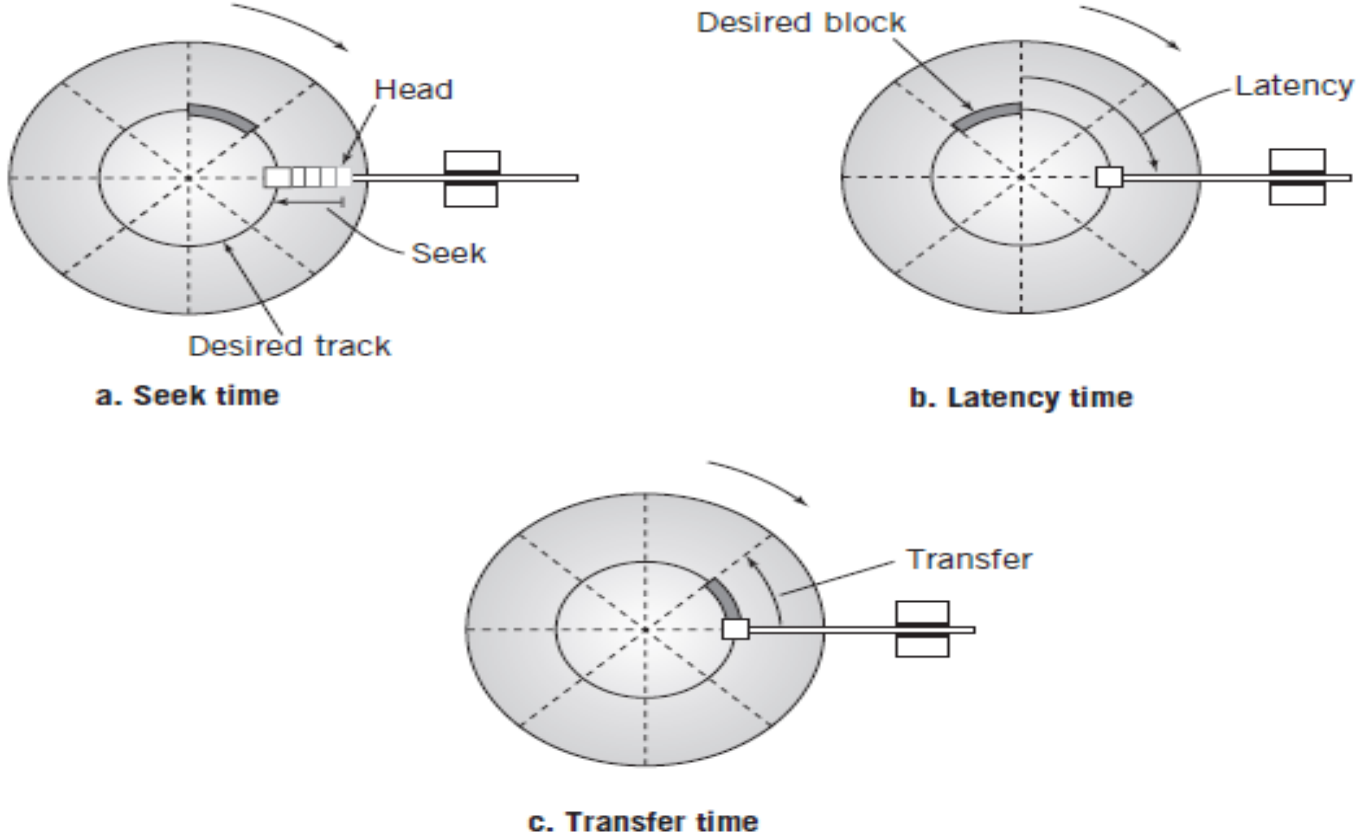


Hard Disk Drive(Cont..)

1. **Seek time –** The time taken by the R-W head to reach the desired track from its current position.
2. **Rotational latency –** Time is taken by the sector to come under the R-W head.
3. **Data transfer time –** Time is taken to transfer the required amount of data. It depends upon the rotational speed.
4. **Controller time –** The processing time taken by the controller.
5. **Average Access time –** seek time + Average Rotational latency + data transfer time + controller time.

Hard Disk Drive(Cont..)

Locating a Block of Data: (a) Seek Time, (b) Latency Time, (c) Transfer Time



Hard Disk Drive(Cont..)

$$\text{average latency} = \frac{1}{2} \times \frac{1}{\text{rotational speed}}$$

- For a typical hard disk rotating at 3600 revolutions per minute, or 60 revolutions per second, the average latency is,

$$\text{average latency} = \frac{1}{2} \times \frac{1}{60} = 8.33\ milliseconds$$

Hard Disk Drive(Cont..)

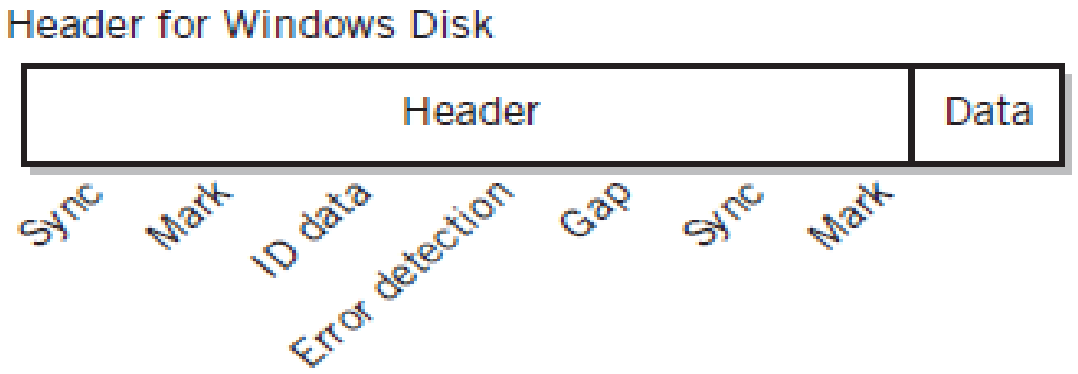
$$\text{transfer time} = \frac{1}{\text{number of sectors} \times \text{rotational speed}}$$

- If the hard drive in the example contains 30 sectors per track, the transfer time for a single block would be

$$\text{transfer time} = \frac{1}{30 \times 60} = 0.55\ milliseconds$$

Hard Disk Drive(Cont..)

- Figure shows the layout of the header for a Windows-based disk.
- The track positions, blocks, and headers must be established before the disk can be used. The process to do this is known as **formatting** the disk.
- The header identifier must be a unique pattern of 1s and 0s, the data being stored must be checked by the disk controller to assure that the data pattern does not accidentally match the header identifier.
- If it does, the pattern stored on the disk is modified in a known way.



Disk Arrays

- It is common to group multiple disks together. Such a grouping of two or more disk drives is called a **disk array** or a **drive array**.
- A disk array can be used to reduce overall data access time by sharing the data among multiple disks and also to increase system reliability by providing storage redundancy.
- One useful type of disk array is known as **RAID**, which stands for **Redundant Array of Inexpensive Disks**.
- There are two standard methods of implementing a disk array. One is known as a **mirrored array**, and the other as a **striped array**.
- A mirrored array consists of two or more disk drives. In a mirrored array, each disk stores exactly the same data.
- During reads, alternate blocks of the data are read from different drives, then combined to reassemble the original data.

Disk Arrays(Cont..)

- The access time for a multiblock read is reduced approximately by a factor equal to the number of disk drives in the array.
- If a read failure occurs in one of the drives, the data can be read from another drive and the bad block marked to prevent future use of that block, increasing system reliability.
- When three drives are used, errors that are not detected by normal read failures can be found using a method known as **majority logic**.
- In a **striped array**, a file segment to be stored is divided into blocks. Different blocks are then written simultaneously to different disks.
- This effectively multiplies the throughput rate by the number of data disks in the array.
- A striped array requires a minimum of three disk drives; in the simplest configuration, one disk drive is reserved for error checking.

Disk Arrays(Cont..)

- As the write operation is taking place, the system creates a block of parity words from each group of data blocks and stores that on the reserved disk.
- During read operations, the parity data is used to check the original data.
- There are five well-defined RAID standards, labeled RAID 1 through RAID 5.
- RAID 1 is a mirrored array RAID 1 provides protection by storing everything at least twice, but offers a substantial performance gain, particularly under heavy data read usage.
- RAIDs 2, 3, and 4 are arrays that are striped in different ways. Each uses a separate disk for error checking.
- RAID 0 is not a true RAID, because it provides no redundancy and no inherent error checking.

Solid-State Drive (SSD)

- SSD is a non-volatile storage device, which stands for Solid State Drive
- Solid-state drive (SSD) is a solid-state storage device that uses integrated circuit assemblies as memory to store data.
- There are no moving mechanical components in SSD.
- This makes them different from conventional electromechanical drives such as Hard Disk Drives (HDDs) or floppy disks, which contain movable read/write heads and spinning disks.

Solid-State Drive (Cont..)

- It is a type of **non-volatile** memory that retains data even when power is lost.
- SSDs may be constructed from random-access memory (RAM) for applications requiring fast access but not necessarily data persistence after power loss.

Features of SSD

- **Start-up Time:** SSDs do not use any mechanical component hence it takes almost negligible startup time.
- **Random Access Times:** Accessing data directly from the Flash memory creates a lag-free experience for its users.
- **Data transfer rates:** Higher Data Transfer rates of about 100-600 Mb/sec.
- **Noise:** As they have no mechanical part they create zero noise.
- **Reliability:** SSDs are reliable as there are no moving parts that can wear over time.
- **Security features:** SSDs offer advanced security features like hardware encryption and secure erase functions to protect data from unauthorized access.

Benefits of using SSD

- **Faster Read/Write Speeds:** SSDs offer significantly faster read and write speeds than traditional hard disk drives (HDDs), which use spinning disks and mechanical read/write heads. This translates into faster boot times, application loading times, and overall system performance.
- **Lower Power Consumption:** SSDs consume less power than HDDs, making them ideal for use in laptops, tablets, and other mobile devices that rely on battery power.
- **No Moving Parts:** Unlike Hard Disk Drives, which have spinning disks and mechanical read/write heads, SSDs have no moving parts. This makes them more resistant to shock and vibration, and less prone to mechanical failure.
- **Higher Reliability:** Because SSDs have no moving parts, they are generally more reliable than HDDs, which are subject to mechanical wear and tear. SSDs also have a lower rate of data loss due to read/write errors.
- **Lower Noise and Heat Output:** SSDs generate less noise and heat than HDDs, making them ideal for use in quiet environments or systems with limited cooling

Disadvantages of SSD

- **Cost:** SSDs are more expensive than regular hard drives.
- **Limited Lifespan:** Each memory cell in an SSD has a limited number of write cycles before it degrades.
- **Storage choices:** Because of the expense, SSDs are often sold in smaller sizes.

Difference Between SSD and HDD

Parameter	HDD	SSD
Long term storage	HDDs are more reliable for long-term storage.	SSDs are comparatively less reliable for long-term storage due to data leaks that can occur if kept unpowered for more than a year.
Access speed	The data accessing speed is slower as compared to SSD.	The data accessing speed is much higher as compared to HDD.
Performance	The performance suffers because of fragmentation.	The performance does not suffer because of fragmentation.
Suitable for	HDDs are suitable for •Extensive storage •Long-term storage	SSDs are suitable for •Fast data retrieval •Laptop or desktop because of low power consumption and size.

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Optical drives

- An optical disc is an electronic data storage medium that is also referred to as an optical disk, optical storage, optical media, Optical disc drive, disc drive, which reads and writes data by using optical storage techniques and technology.
- An optical disc, which may be used as a portable and secondary storage device, was first developed in the late 1960s. James T. Russell invented the first optical disc, which could store data as micron-sized light and dark dots.
- An optical disc can store more data and has a longer lifespan than the preceding generation of magnetic storage medium.

Optical drives(Cont..)

- To read and write to CDs and DVDs, computers use a CD writer or DVD writer drive, and to read and write to Blu-ray discs, they require a Blu-ray drive.
- The CDs, Blu-ray, and DVDs are the most common types of optical media, which are usually used to:

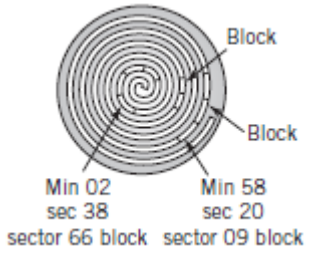
1. They are used to transfer data to various devices or computers.
2. These media are used to deliver the software to others.
3. They help users to hold large amounts of data, like videos, photos, music, and more.
4. Also, optical media are used to get back up from a local machine.

CD-ROM

- CD-ROM, short for compact disk read-only memory, was the first disk on the basis of drives for the latest PCs.
- To reduce noise and increase stability, most CD-ROM drives in computers run at a slower speed, and if the drive experiences read errors, it will only speed up for larger data files.
- The newest CD-ROM drives have the potential to achieve read speeds of 60 revolutions in a second (60x).
- **CD-ROM** data storage is similar to magnetic disk: data is stored in blocks on the disk. The blocks can be arranged in files, with a directory structure similar to that of magnetic disks.
- Rather than concentric tracks, data on a CD-ROM is stored on a single track, approximately three miles long.

CD-ROM(Cont..)

- A CD-ROM typically stores 270,000 blocks of data. Each block is 2352 bytes long and
- holds 2048 bytes of data.
- Due to the difficulty of the manufacturing process, errors can occur, so the CD-ROM provides extensive means for correcting the errors.
- Blocks on a CD-ROM are identified by a 4-byte identification code that was inherited from the audio origins of the medium. Three bytes, stored in binary-coded decimal (BCD) format, identify the block by minute, second, and sector.
- The fourth byte identifies a mode of operation. Mode 1, the normal data mode, with error correction. Mode 2 increases the capacity by eliminating the error correction. Other modes are provided for special audio and video features.



DVD-ROM

- DVD-ROM drives, which stand for Digital Versatile Disk Read Only Memory and are a direct evolution from CD-ROM drives, have significantly more performance and capacity than their CD counterparts while maintaining the same physical dimensions.
- **DVD** technology is essentially similar to CD-ROM technology. The disk is the same size, and is formatted similarly.
- The use of a laser with a shorter light wavelength (visible red, instead of infrared) allows tighter packing of the disk In addition, the laser can be focused in such a way that two layers of data can be placed on the same side of the disk, one underneath the other.
- Each layer on a DVD can hold approximately 4.7 GB. If both layers on both sides are used, the DVD capacity is approximately 17 GB.
- The use of a blue laser extends this capability even further, to approximately 50 GB.

Blu-ray

- Blu-Ray technology squashes even more data into the same size disc as a CD or DVD.
- **Disc capacities**

CD	DVD	Blu-Ray
700 MB	4.7 GB	25 GB – 128 GB

WORM or write-once-read-many-times

- **WORM**, or **write-once-read-many-times**, disks were originally designed to provide an inexpensive way for archiving data.
- WORM disks provide high-capacity storage with the convenience of compact size, reasonable cost, and removability.
- As the name indicates, WORM disks can be written, but, once written, a data block cannot be rewritten.
- WORM disks work similarly to a CD or DVD. The major difference is that the disk is made of a material that can be blistered by a medium-power laser.
- Initially, the entire disk is smooth. When data is to be written, the medium-power laser creates tiny blisters in the appropriate locations.
- These correspond to the pits in a normal CD-ROM. The WORM disk is read with a separate low-power laser in the same way as a CD-ROM.

Advantages of Optical Disk

- **Cost :** Only plastics and aluminum foils are used in the production of an optical disk, which makes their manufacturing cost less expensive.
- **Durability :** While comparing with Volatile and Non-Volatile memories, optical disks are more durable. It is not caused to data losses due to any power failure and is not subjected to wear.
- **Simplicity:** With the help of using optical disks, the process of backup of data is much easier.
- **Stability :** A very high level of stability is provided by optical disk because it is not unprotected from electromagnetic fields and other kinds of environmental influences, unlike magnetic disks.
- **Portability :** Optical disks are very portable; however, they are fairly large in size. They can be used in different computers and devices and transported to different places as they can be placed inside bags and other small objects.

I/O Communications

- **I/O (Input/Output) communication** refers to the process by which a computer exchanges data with external devices, such as keyboards, monitors, storage devices, and network interfaces.
- It enables interaction between the computer's CPU, memory, and peripheral devices.

Types of I/O Communication:

1. **Input Devices**: Devices that send data to the computer.
    1. Examples: Keyboard, Mouse, Microphone, Scanner.
2. **Output Devices**: Devices that receive data from the computer.
    1. Examples: Monitor, Printer, Speakers.
3. **Storage Devices**: Used for data storage and retrieval.
    1. Examples: Hard drives, SSDs, USB drives.
4. **Network Devices**: Enable data transfer over a network.
    1. Examples: Network Interface Cards (NICs), Routers.

Methods of I/O Communication:

- **Memory-mapped I/O**
- **Direct memory access (DMA)**
- **Polling I/O**
- **Interrupts I/O**

Memory-mapped I/O:

- When using memory-mapped I/O, the same address space is shared by memory and I/O devices.
- The device is connected directly to certain main memory locations so that I/O device can transfer block of data to/from memory without going through CPU.

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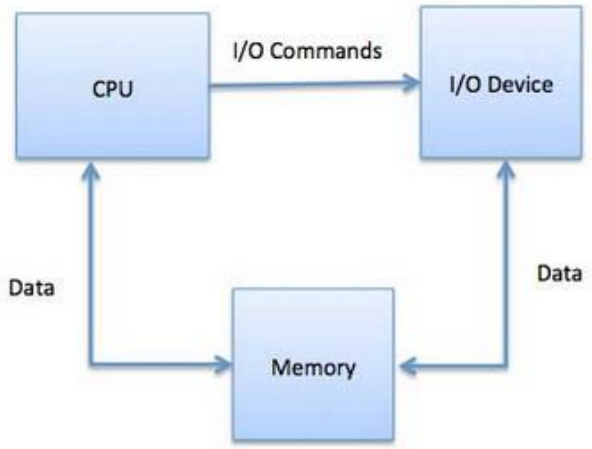
- While using memory mapped IO, OS allocates buffer in memory and informs I/O device to use that buffer to send data to the CPU.
- The advantage to this method is that every instruction which can access memory can be used to manipulate an I/O device.
- Memory mapped IO is used for most high-speed I/O devices like disks, communication interfaces.



Direct Memory Access (DMA)

- Direct Memory Access (DMA) means CPU grants I/O module authority to read from or write to memory without involvement.
- DMA module itself controls exchange of data between main memory and the I/O device.
- CPU is only involved at the beginning and end of the transfer and interrupted only after entire block has been transferred.
- Direct Memory Access needs a special hardware called DMA controller (DMAC) that manages the data transfers and arbitrates access to the system bus.

Advantages of DMA:

- **Frees CPU Resources**: Allows the CPU to perform other tasks while data is being transferred.
- **Faster Data Transfers**: Bypasses the need for CPU intervention.
- **Efficient for Large Data Volumes**: Reduces overhead for bulk transfers.

Disadvantages of DMA:

- **Hardware Complexity**: Requires a dedicated DMA controller.
- **Memory Contention**: Can cause delays if both the CPU and DMA controller need memory access simultaneously.
- **Software Overhead**: The CPU still needs to configure and manage the DMA controller.

Polling I/O

- Polling is the simplest way for an I/O device to communicate with the processor.
- The process of periodically checking status of the device to see if it is time for the next I/O operation, is called polling.
- The I/O device simply puts the information in a Status register, and the processor must come and get the information.
- Most of the time, devices will not require attention and when one does it will have to wait until it is next interrogated by the polling program. This is an inefficient method.

Interrupts I/O

- An alternative scheme for dealing with I/O is the interrupt-driven method.
- An interrupt is a signal to the microprocessor from a device that requires attention.
- A device controller puts an interrupt signal on the bus when it needs CPU's attention when CPU receives an interrupt'
- It saves its current state and invokes the appropriate interrupt handler using the interrupt vector (addresses of OS routines to handle various events).
- When the interrupting device has been dealt with, the CPU continues with its original task.

Device Management

- **Device Management** is a core function of an operating system (OS) that handles the interaction and control of hardware devices connected to a computer system.
- It ensures that devices like printers, storage drives, and network interfaces work efficiently and are accessible to applications and users.

Types of Devices Managed: Input Devices, Output Devices , Storage Devices, Communication Devices etc

Functions of the Device Management

1. The device management system works as the **Input/Output controller** which means it keeps track of all the device's data, location, status, etc.
2. The device management system in the Operating system **allocates and deallocates the devices**.
3. The device management system in the Operating system **creates schedules and decides on the basis of priority which process will hold a particular device and for how long it will hold it**.

Interface cards

- **Interface cards** (also called expansion cards or adapter cards) are hardware components inserted into a computer's motherboard to expand its functionality and enable communication with peripheral devices.
- **Types of Interface Cards:**

1. **Network Interface Card (NIC)**:
2. **Graphics Card (GPU)**:
3. **Sound Card**:

Network Interface Cards

- A Network Interface Card (NIC) is an essential component of any computer that connects it to the network.
- It **enables data to be sent and received between the computer and other network-connected devices, such as routers or switches**.
- A NIC provides a computer with a dedicated, full-time connection to a network.
- The NIC operates as a middleman between a computer and a data network.
- For example, when a user requests a webpage, the computer passes the request to the network card, which converts it into electrical impulses.
- A web server on the internet receives the impulses and responds by sending the webpage back to the network card as electrical signals.

- The card receives these signals and translates them into the data that the computer displays.
- The term *network interface card* is often interchangeable with the terms *network interface controller*, *network adapter* and *LAN adapter*.
- Network controllers were originally implemented as expansion cards that could plug into a computer port, router or USB device.
- However, more modern controllers are built directly into the computer motherboard chipset.

Types of NICs

- **Wireless.** NICs that use an antenna to provide wireless reception through radio frequency waves. Wi-Fi connections use wireless NICs.
- **Wired.** NICs that have input jacks made for cables. Ethernet is the most popular wired LAN technology.
- **USB.** NICs that provide network connections through a device plugged into the USB port.
- **Fiber optics.** NICs used as a high-speed support system for network traffic handling on server computers. It's also possible to achieve this support by combining multiple NICs. Fiber optic NICs are typically more expensive and complex.

NIC components

- **Speed.** All NICs have a speed rating in terms of megabits per second (Mbps) that determines the card's performance in a network. The average Ethernet NICs come in 10 Mbps, 100 Mbps, 1000 Mbps and 1 gigabits per second varieties.
- **Driver.** The required software that passes data between the computer's operating system and the NIC. When a NIC is installed on a computer, the corresponding driver software is also downloaded.
- **MAC address.** Unique, unchangeable media access control addresses, also known as physical network addresses, are assigned to NICs.
- **LED indicator.** Most NICs have an LED indicator integrated into the connector to notify the user when the network connects and data transmission occurs.
- **Router.** A router is sometimes needed to enable communication between a computer and other devices. In this case, the NIC connects to the router which is connected to the internet.

Graphics Card

- GPU stands for Graphics Processing Unit. The power of GPU depend on the model of the GPU.
- The graphics as an external component is attached on a slot known as an expansion slot. It is the brain of the graphics card and is what creates the visuals that we see on the monitor.
- **Types of Graphics Card**

1. **Integrated –** The graphics which are built into the motherboard are known as Integrated, are generally used in most laptops, the cannot be easily upgraded.

2. **Discrete –** It is an external graphics card which is a hardware and added on a motherboard as an extra component.

Features of Graphics Card

- **Memory:** Graphics card carries its own memory. Memory range could be from 128MB to 2GB of memory.
- **Multiple Screen support:** Most new video cards have the ability to connect two monitors to one card.
- **Connection –** The graphic card is connected to the monitor using many different ports put the port must be present on both monitor and Graphics card.
- These are some common ports used to connect graphics card with a monitor.
- VGA
- HDMI
- DVI

Sound Cards

- The sound card is a part of a computer that produces and records sound. Users may use it to connect analogue microphones, speakers, and headphones to their computers.
- Usually connected via USB, external sound cards are also known as audio interfaces.

Uses of Sound Card

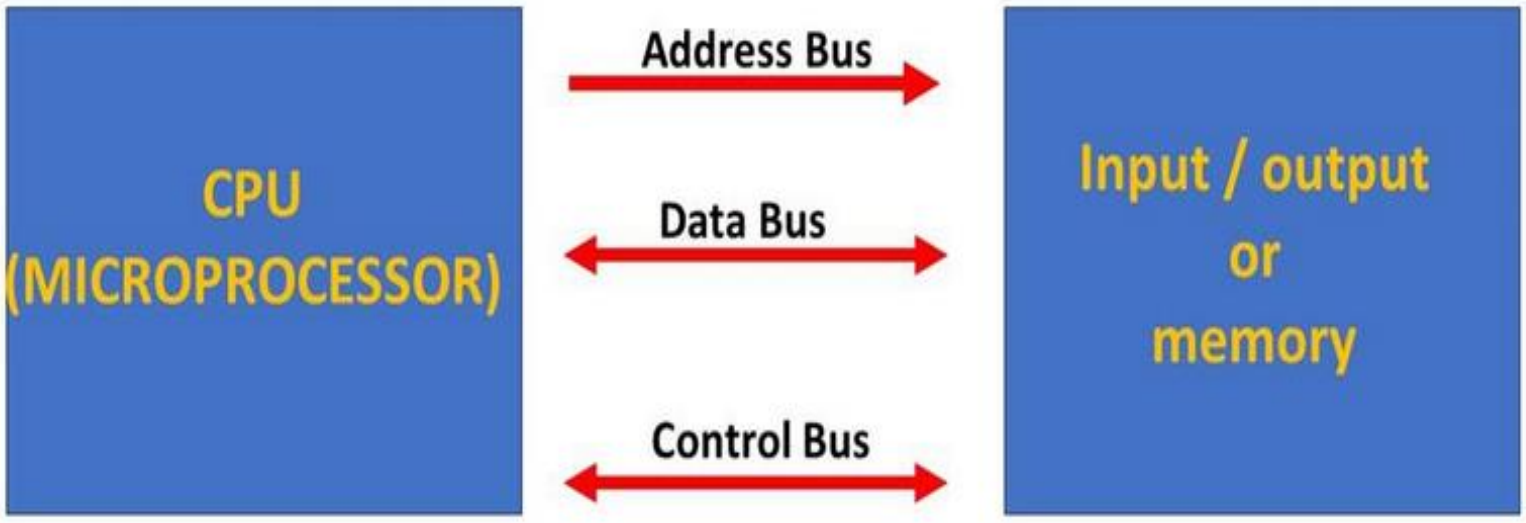
- **Audio recording:** The card has a microphone input port that may be used to record sounds such as voices, instruments, and other noises.
- **Production of music:** Recording studios can enhance their output by utilizing external sound cards.
- **Sound for gaming:** Users may also utilize an external sound card to enhance surround sound and positional audio for the greatest gaming experiences.

BUSES in computer

- A computer bus is a communication system within a computer or between computers that transfers data between different components.
- The purpose of buses is to reduce the number of "pathways" needed for communication between the components, by carrying out all communications over a single data channel.
- A bus is a set of physical connections (cables, circuits, etc.) that can be shared by multiple hardware components to communicate with one another.
- Memory and input/ output devices are connected to the Central Processing Unit through a group of lines called a bus.

Types of Computer Bus

1. Address Bus
2. Data Bus
3. Control Bus



Computer Bus

Address Bus

- A collection of wires used to identify particular location in main memory is called Address Bus.
- The address bus transports memory addresses which the processor wants to access in order to read or write data.
- The address bus is **unidirectional**.
- The size of address bus determines how many unique memory locations can be addressed.
- A system with 4-bit address bus can address 2^4 = 16 Bytes of memory.

Data Bus

- A collection of wires through which data is transmitted from one part of a computer to another is called Data Bus.
- Data Bus can be thought of as a highway on which data travels within a computer.
- The main objective of data bus is transfer of the data between microprocessor to input/ output devices or memory.
- The data bus transfers instructions coming from or going to the processor.
- The data bus is **bidirectional** because the data can flow in either direction from CPU to memory(or input/output device) or from memory to the CPU.
- The size (width) of bus determines how much data can be transmitted at one time.

Example:

- A 16-bit bus can transmit 16 bits of data at a time.
- 32-bit bus can transmit 32 bits at a time.

Control Bus

- The connections that carry control information between the CPU and other devices within the computer is called Control Bus.
- The main objective of control bus is all signals controller carried from processor to other hardware device.
- The Control bus is **bidirectional** because the data can flow in either direction from CPU to memory(or input/output device) or from memory to the CPU.
- It also transmits response signals from the hardware.

Firmware

- Firmware is a form of microcode or program embedded into hardware devices to help them operate effectively.
- Hardware like cameras, mobile phones, network cards, optical drives, printers, routers, scanners, and television remotes rely on firmware built into their memory to function smoothly.
- Firmware is often referred to as “software for hardware.”
- Firmware provides instructions to help hardware start up, communicate with other devices, and perform basic input/output tasks.
- Software, on the other hand, is installed onto a device and used for interaction, such as browsing the internet, word processing, listening to music, and videoconferencing.

- Hardware manufacturers regularly release firmware updates so their devices remain secure and compatible with new media.
- For example, Android and iOS devices periodically get firmware updates that enhance their performance, add new features, and protect them against security threats and vulnerabilities.

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Types And Examples Of Firmware

- When a device switches on, the firmware instructs the processor to initiate the startup process.
- There are typically three levels of firmware:
- **Low-level firmware**: This firmware is usually stored in non-volatile memory chips like read-only memory (ROM) and one-time programmable (OTP) memory. These chips cannot be rewritten or updated, and the firmware is intrinsic to the hardware, such as a computer.
- **High-level firmware**: This firmware is deployed within flash memory chips and comes with more complex instructions that allow updates to be made.
- **Subsystems**: These are semi-independent devices that are part of a more extensive system. Firmware at this level is embedded within central processing units (CPUs), flash chips, and liquid crystal display (LCD) units.

Common types of firmware include:

- **1. BIOS :** The Basic Input/Output System (BIOS) has been a critical component of computers since the very early days.
- The BIOS sits on a chip in a computer's motherboard and issues a set of instructions that enables the device's operating system to load.
- It is also responsible for looking after a device's hardware components and ensuring they function correctly upon startup.
- When a computer is turned on, the BIOS launches instructions that check the machine's startup process for potential errors.
- It starts by checking the random access memory (RAM) and processor for faults, then checks connected devices, such as a keyboard and mouse, for any problems.

- It then checks for a boot sequence, such as Boot from Compact Disc Read-Only Memory (CD-ROM) and Boot from Hard Drive.
- Finally, the BIOS connects to the bootloader program, which wakes up the computer's operating system and loads it in the RAM.

2. EFI : Extensible Firmware Interface (EFI) is a new generation of firmware that issues the instructions that a device's CPU uses to start the hardware and initiate the bootloader.

- EFI is a replacement for BIOS and is used in new computers.
- The original EFI firmware is used in Macs and can only support Intel processors.
- Unified EFI (UEFI) is a piece of firmware designed to replace the BIOS in new Windows computers and supports AMD and Intel processors.

Boot process

- The procedure of starting a computer by loading the kernel is known as **booting** the system.
- On most computer systems, a small piece of code known as the **bootstrap program** or **bootstrap loader** locates the kernel, loads it into main memory, and starts its execution.
- Some computer systems, such as PCs, use a two-step process in which a simple bootstrap loader fetches a more complex boot program from disk, which in turn loads the kernel.
- When a CPU receives a reset event—for instance, when it is powered up or rebooted—the instruction register is loaded with a predefined memory location, and execution starts there.
- At that location is the initial bootstrap program. This program is in the form of **read-only memory (ROM).**
- ROM is convenient because it needs no initialization and cannot easily be infected by a computer virus.

Types of Booting

1. Cold Booting : A cold boot is also called **a hard boot.**

- It is the process when we first start the computer. When the computer is started from its initial state by pressing the power button it is called cold boot.
- The instructions are read from the ROM and the operating system is loaded in the main memory.

2. Warm Booting : Warm Boot is also called **soft boot**. It refers to when we restart the computer.

- When the system gets stuck sometimes it is required to restart it while it is ON. Therefore, in this condition the warm boot takes place.
- Restart button or CTRL+ALT+DELETE keys are used for warm boot.

Steps of Boot Process

1. **The Startup :** It is the first step that involves switching the power ON. It supplies electricity to the main components like BIOS and processor.
2. **BIOS: Power On Self Test :** It is an initial test performed by the BIOS. Further, this test performs an initial check on the input/output devices, computer's main memory, disk drives, etc. Moreover, if any error occurs, the system produces a beep sound.
3. **Loading of OS:** In this step, the operating system is loaded into the main memory. The operating system starts working and executes all the initial files and instructions.
4. **System Configuration:** In this step, the drivers are loaded into the main memory. Drivers are programs that help in the functioning of the peripheral devices.
5. **Loading System Utilities :**System utilities are basic functioning programs, for example, volume control, antivirus, etc. In this step, system utilities are loaded into the memory.

6. **User Authentication :** If any password has been set up in the computer system, the system checks for user authentication. Once the user enters the login Id and password correctly the system finally starts.

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